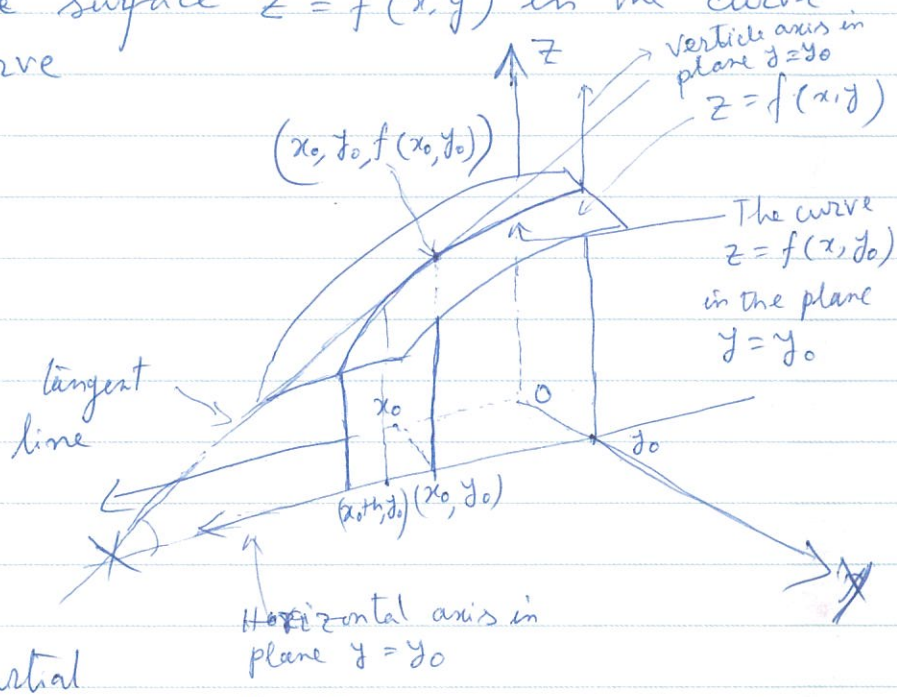
Partial Derivatives

(of a function of two variables)

If (x_0, y_0) is a point in the domain of $f(x,y)$, then the vertical plane $y = y_0$ will cut the surface $z = f(x,y)$ in the curve $z = f(x, y_0)$. This curve is the graph of the function $z = f(x, y_0)$ in the plane $y = y_0$. The horizontal coordinate in this plane is x; and the vertical coordinate is z. The y value is held constant at y_0, so y is not a variable.



We define the partial derivative of f w.r.to x at the pt. (x_0, y_0) as the ordinary derivative of $f(x, y_0)$ w.r.to x at the pt. $\cancel{x = y_0}$ $x = x_0$.

and which is the slope of the tangent to the curve $f(x, y_0)$ at x_0.

<u>Def</u> The partial derivative of $f(x,y)$ with respect to x at the point (x_0, y_0) is

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)} = \lim_{h \to 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

$$\therefore \left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)} = \left.\frac{d}{dx} f(x, y_0)\right|_{x = x_0}$$

<u>Note</u>: $\left.\frac{\partial f}{\partial x}\right|_{(x_0, y_0)} = f_x(x_0, y_0) = \left.\frac{\partial z}{\partial x}\right|_{(x_0, y_0)}$

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Similarly: The partial derivative of $f(x,y)$ with respect to y at the point (x_0, y_0) is

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy} f(x_0, y)\right|_{y=y_0} = \lim_{h\to 0} \frac{f(x_0, y_0+h) - f(x_0, y_0)}{h}$$

Find first order P.D. of

<u>Ex-1</u> $f(x,y) = \dfrac{x+y-1}{x-y+1}$ at $(2,1)$ using def.

<u>Sol^n.</u>: $$\left.\frac{\partial f}{\partial x}\right|_{(2,1)} = \lim_{h\to 0} \frac{f(2+h,1) - f(2,1)}{h} \cancel{=} \cancel{(2+h)}$$

$$= \lim_{h\to 0} \frac{\frac{2+h}{2+h} - \frac{2}{2}}{h} = \lim_{h\to 0} \frac{1-1}{h} = 0$$

$$\left.\frac{\partial f}{\partial y}\right|_{(2,1)} = \lim_{h\to 0} \frac{f(2, 1+h) - f(2,1)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{2+h}{2-h} - \frac{2}{2}}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{2}+h-\cancel{2}+h}{h(2-h)} = \lim_{h\to 0} \frac{2}{2-h} = 1$$

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the values of

Ex-2 Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(4,-5)$ where

$$f(x,y) = x^2 + 3xy + y - 1$$

Solⁿ. To find $\frac{\partial f}{\partial x}$ we treat y as constant and differentiate f w.r.to x:

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(x^2 + 3xy + y - 1\right)$$

$$= 2x + 3y + 0 - 0$$

$$\therefore \left.\frac{\partial f}{\partial x}\right|_{(4,-5)} = 2(4) + 3(-5) = -7$$

Similarly $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(x^2 + 3xy + y - 1\right)$

$$= 0 + 3x + 1 - 0$$

$$\therefore \left.\frac{\partial f}{\partial y}\right|_{(4,-5)} = 3(4) + 1 = 13$$

Ex-3: Fin f_x and f_y of the function

$$f(x,y) = \frac{2y}{y + \cos x}$$

Solⁿ: $f_x = \frac{\partial}{\partial x}\left(\frac{2y}{y + \cos x}\right)$

$$= \frac{(y + \cos x)\frac{\partial}{\partial x}(2y) - (2y)\frac{\partial}{\partial x}(y + \cos x)}{(y + \cos x)^2}$$

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$$= \frac{(y+\cos x)(0) - (2y)(-\sin x)}{(y+\cos x)^2} = \frac{2y\sin x}{(y+\cos x)^2}$$

$$f_y = \frac{\partial}{\partial y}\left(\frac{2y}{y+\cos x}\right)$$

$$= \frac{(y+\cos x)\frac{\partial}{\partial y}(2y) - (2y)\frac{\partial}{\partial y}(y+\cos x)}{(y+\cos x)^2}$$

$$= \frac{(y+\cos x)(2) - (2y)(1)}{(y+\cos x)^2} = \frac{2\cos x}{(y+\cos x)^2}$$

<u>Ex-4</u> Find $\frac{\partial z}{\partial x}$ assuming that the eqn. $yz - \ln z = x + y$ defines z as fn. of two independent variables x and y and z_x exists.

<u>Soln</u>: We differentiates both sides of the eqn. w.r.to x, holding y constant and treating z as a differentiable fn. of x:

$$\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial x}(\ln z) = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial x}(y)$$

$$\Rightarrow \left\{z\frac{\partial}{\partial x}(y) + y\frac{\partial}{\partial x}(z)\right\} - \frac{1}{z}\frac{\partial}{\partial x}(z) = 1 + 0$$

$$\Rightarrow 0 + y.z_x - \frac{1}{z}z_x = 1$$

$$\Rightarrow \left(y - \frac{1}{z}\right)z_x = 1$$

$$\Rightarrow z_x = \frac{z}{yz-1}$$

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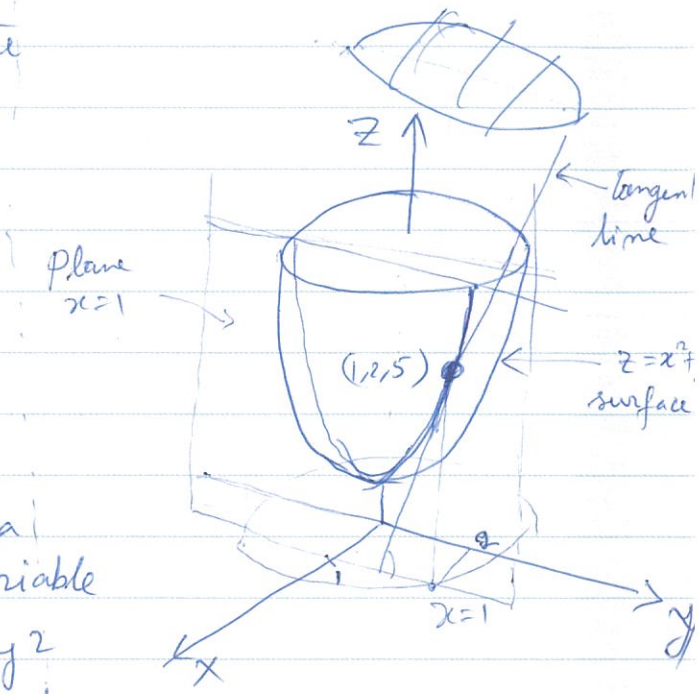
Ex-5 The plane $x=1$ intersects the paraboloid $z=x^2+y^2$ in a parabola. Find the ~~sp~~ slope of the tangent to the parabola at $(1,2,5)$.

Sol^n: The parabola lies in ~~the~~ plane parallel to the y-z plane, and the slope is the value of the partial derivative $\left.\frac{\partial z}{\partial y}\right|_{(1,2)}$

$$= \left.\frac{\partial}{\partial y}\left(x^2+y^2\right)\right|_{(1,2)}$$

$$= \left.(2y)\right|_{(1,2)} = 4$$



Check: We can treat the parabola as the graph of the single variable function $z=(1)^2+y^2=1+y^2$ in the plane $x=1$ and find the slope at $y=2$, i.e.,

$$\left.\frac{dz}{dy}\right|_{y=2} = \left.\frac{d}{dy}\left(1+y^2\right)\right|_{y=2} = \left.(2y)\right|_{y=2} = 4$$

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2nd Order Partial Derivatives

(Partially) Differentiate $f(x, y)$ ~~to~~ twice to get a second order (Partial) derivative.

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) \equiv \frac{\partial^2 f}{\partial x^2} \equiv f_{xx}$$

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) \equiv \frac{\partial^2 f}{\partial x \partial y} \equiv f_{xy}$$

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) \equiv \frac{\partial^2 f}{\partial y \partial x} \equiv f_{yx}$$

Mixed second order Partial derivatives.

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) \equiv \frac{\partial^2}{\partial y^2} \equiv f_{yy}$$

Ex-1 $\quad f(x, y) = x\cos y + y e^x$

$\because \dfrac{\partial f}{\partial x} = \cos y + y e^x$ & $\dfrac{\partial f}{\partial y} = -x\sin y + e^x$

$\therefore \dfrac{\partial^2 f}{\partial x^2} = y e^x$ $\quad \dfrac{\partial^2 f}{\partial y^2} = -x\cos y$

$\dfrac{\partial^2 f}{\partial y \partial x} = -\sin y + e^x$ $\quad \dfrac{\partial^2 f}{\partial x \partial y} = -\sin y + e^x$

Here $f_{xy} = f_{yx}$ but it is not the case always.

Ex-2: $f(x, y) = xy + \dfrac{e^y}{y^2+1}$

$f_x = y$ $\qquad f_y =$

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<u>Mixed Derivative Theorem</u>: If $f(x,y)$ and its partial derivatives f_x, f_y, f_{xy}, f_{yx} are define throughout an open region containing a point $(x_0\ y_0)$ and all are continuous at (x_0, y_0) then $f_{xy}(x_0, y_0) = f_{yx}(x_0, y_0)$.

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The Chain Rule

Function of one variable: $w = f(x)$ is a differentiable fn. of x and $x = g(t)$ is a differentiable fun of $t \Rightarrow w$ is differentiable fn of t and

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt}$$

Function of two variables:

With One independent variable (two intermediate var.): If $w = f(x, y)$ is differentiable and if $x = x(t)$, $y = y(t)$ are differentiable funs of t then $w = f(x(t), y(t))$ is differentiable fun of t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

Ex:- Use the Chain Rule to find the derivative of $w = xy$ w.r.to t along the path $x = \cos t$, $y = \sin t$. What is the derivative's value at $t = \pi/2$? Also find $\frac{dw}{dt}$ by direct method

Solⁿ:- We apply Chain Rule to find $\frac{dw}{dt}$ as follows:

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$$

$$= \frac{\partial (xy)}{\partial x} \cdot \frac{d}{dt}(\cos t) + \frac{\partial (xy)}{\partial y} \cdot \frac{d}{dt}(\sin t)$$

$$= (y)(-\sin t) + (x)(\cos t)$$

$$= (\sin t)(-\sin t) + (\cos t)(\cos t)$$

$$= \cos^2 t - \sin^2 t = \cos 2t$$

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Direct method: $w = xy = \cos t \cdot \sin t =$

$$= \frac{1}{2}\sin 2t$$

$$\therefore \frac{dw}{dt} = \frac{d}{dt}\left(\frac{1}{2}\sin 2t\right)$$

$$= \frac{1}{2}(2\cos 2t) = \cos 2t$$

Now $\left.\frac{dw}{dt}\right|_{t=\pi/2} = \cos\left(2 \cdot \pi/2\right) = \cos\pi = -1$

Two independent variables and two intermediate vars. :-

If $w = f(x, y)$, $x = g(s, t)$, $y = h(s, t)$ are differentiable fns then w has partial derivatives w. r. to s and t, and given by the formulas

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t}.$$

Ex- Express $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$ in terms s and t if $w = x^2 + y^2$ and $x = s - t$ $y = s + t$.

Solⁿ. $\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}$

$$= (2x)(1) + (2y)(1)$$

$$= 2(s-t) + 2(s+t) = 4s$$

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= (2x)(-1) + (2y)(1)$$

$$= -2(s-t) + 2(s+t) = 4t$$

$w = f(x)$

$x = g(s, t)$

$\frac{\partial w}{\partial s} = \frac{dw}{dx} \cdot \frac{\partial x}{\partial s}$

$\frac{\partial w}{\partial t} = \frac{dw}{dx} \cdot \frac{\partial x}{\partial t}$

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